

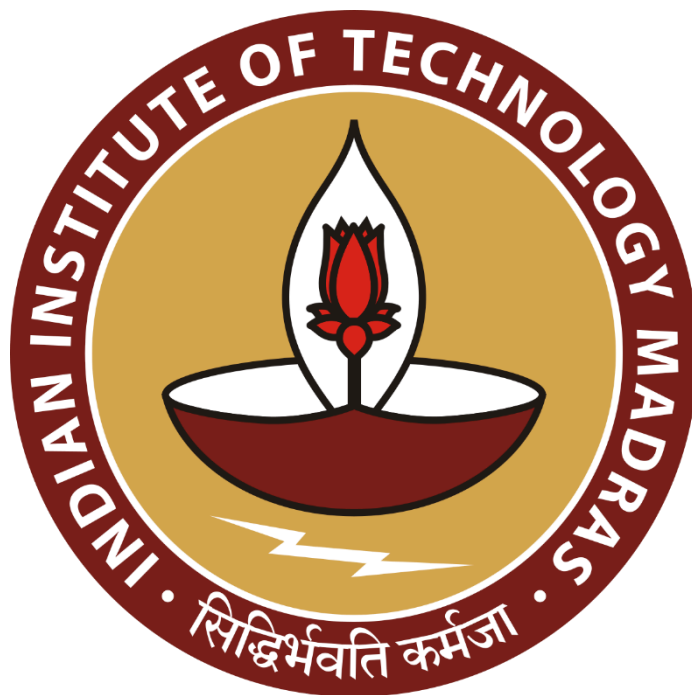
Inventory Management and Purchase Optimization Analysis for Rajesh Surgical

A Final Report for the BDM Capstone Project

Submitted by

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1. Executive Summary

The project titled **“Inventory Management and Purchase Optimization Analysis for Rajesh Surgical”** aims to enhance the company’s operational efficiency and profitability through a data-driven analytical framework. Rajesh Surgical, a Moradabad-based wholesaler of surgical and medical supplies, serves as a crucial intermediary between manufacturers and its network of hospitals, clinics, and retail outlets.

The organization faces inefficiencies in its purchasing and inventory practices, including overstocking of low-demand items that lock up working capital and understocking of high-demand products that lead to missed sales opportunities. Furthermore, existing procurement decisions do not adequately consider price volatility, resulting in missed opportunities for cost savings.

To address these challenges, purchase data from April 2022 to March 2025 was cleaned, standardized, and analyzed using Python and Microsoft Excel. ABC classification was performed to categorize items based on their contribution to total purchase value, followed by VED classification to assess their functional criticality. This dual analysis emphasized the need for tighter control over high-value and high-criticality products that are vital to business operations.

The AV (A-Vital) category items were further classified into regular and intermittent products based on their coefficient of variation (CV) and percentage of zero-demand months, revealing distinct demand patterns. Subsequent forecasting using the Holt-Winters exponential smoothing model (for regular-demand items) and the Croston method (for intermittent-demand items) predicted future requirements for these AV products. Complementary cost optimization analysis identified price-volatile items and optimal purchase months, indicating that the company could potentially achieve 15–30% savings in procurement costs for selected products. Together, these insights provide a comprehensive, data-driven strategy for efficient inventory control and smarter purchase planning at Rajesh Surgical.

2. Detailed Explanation of Analysis Process & Methods

2.1 Data Collection & Data Preprocessing

The raw data was provided directly by Rajesh Surgical as three separate Excel files, covering April 2022–March 2023, April 2023–March 2024, and April 2024–March 2025. Each file contained day-wise purchase transactions. The dataset included the columns Date, Bill, Item Description, Batch, Quantity, Free, Rate, and Amount. For this analysis, the relevant fields were identified as Date, Item Description, Quantity, Rate, and Amount, as they directly influence purchase value, demand, and pricing trends. (Link to [Raw Data](#))

The three files were merged into a single consolidated dataset. Several rounds of data cleaning and preprocessing were performed before analysis. Column names were standardized, and missing or misaligned values were corrected. Misalignment typically occurred when cells shifted left or right during data entry; these were fixed using pattern-based and manual checks. Item descriptions were standardized by converting them to uppercase and removing punctuation, ensuring that identical products with slightly different spellings were treated as one. Duplicate records were dropped, and invalid or summary rows such as “TOTAL” or “GRAND TOTAL” were removed.

Numeric columns (*Quantity*, *Rate*, *Amount*) were verified for consistency, ensuring all entries were numeric. Where possible, the “Amount” column was cross-validated as “Quantity x Rate” to detect anomalies. Items with significant discrepancies were flagged for manual review. The resulting cleaned dataset contained structured day-wise entries across all three financial years and was exported for further analysis in Excel.

(Link to [Cleaned Datasets and the Data Cleaning Code](#))

2.2 Analysis Methods for Identified Problems-

The analysis was designed to address the following two key business problems faced by *Rajesh Surgical*. Let’s focus on Problem Statement-1 first.

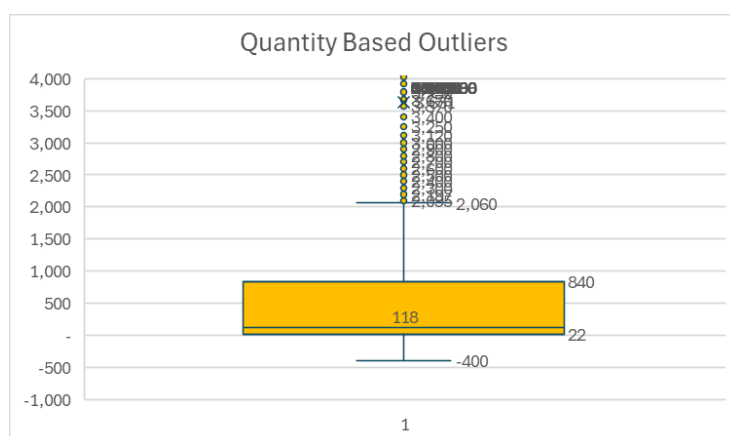
2.2.1 Problem Statement 1: Inventory Planning & Suboptimal Purchase Strategy

2.2.1.1. Objective-

To develop a data-driven inventory management system that enables Rajesh Surgical to maintain optimal stock levels, prevent understocking or overstocking, and forecast future purchase requirements for critical medical supplies.

2.2.1.2 Outliers Detection-

1. Interpretation for Quantity Outliers



(Fig 1- Quantity-Based Outliers)

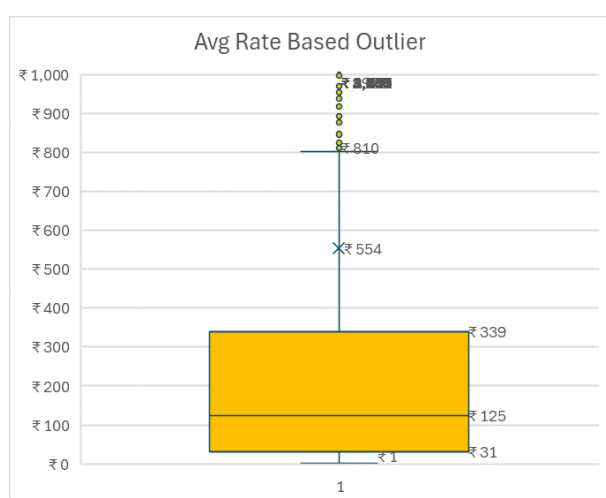
A total of 1,703 items were analyzed based on their total purchased quantity between April 2024 and March 2025. The first quartile (Q1) is 22 units, meaning that 25% of items recorded annual quantities below this level. The median (Q2) is 118 units, indicating that half of all items fall below this value. The third quartile (Q3) is substantially higher at 840 units, reflecting a wide spread in consumption patterns across products. The interquartile range (IQR) therefore stands at 818 units. Applying the standard outlier detection rule ($Q1 - 1.5 \times IQR$ and $Q3 + 1.5 \times IQR$), the lower bound is -1205 units—an impossible value in

Category	Count
Below IQR (Low Outliers)	0
Above IQR (High Outliers)	281
Normal Range	1422

(Table 1: Quantity based Outliers)

real-world data—while the upper bound is 2067 units. Items with total quantities above this upper limit are classified as high-quantity outliers. The outlier analysis based on total quantity indicates that while most items fall within the normal consumption range, 281 products (about 16.5%) exhibit unusually high total purchase quantities. These high outliers likely represent critical consumables, high-turnover items, or bulk-procured products that dominate the purchasing volume. The absence of low outliers is expected because negative quantities are not possible, and very low-demand items remain within the lower normal range. Identifying these high-quantity outliers is valuable for forecasting accuracy, inventory planning, and prioritizing procurement since these items contribute disproportionately to total demand.

2. Interpretation for Rate Outliers



(Fig 2- Rate-Based Outliers)

In Fig. 2, 1,703 items were examined, and the distribution of purchase rates shows substantial variation. The first quartile (Q1) stands at ₹30.85, indicating that 25% of items are priced below this level. The median rate (Q2) is ₹124.74, representing the midpoint of the data, while the third quartile (Q3) is ₹338.74, meaning 75% of items fall below this rate. The resulting interquartile range (IQR) of ₹307.89 reflects a wide spread in pricing across products. Using the standard outlier rule ($Q1 - 1.5 \times IQR$, $Q3 + 1.5 \times IQR$), the lower threshold is ₹-430.99, which is not applicable since rates cannot be negative, and the upper threshold is ₹800.59. Any item with an average rate above this upper bound is considered a high-rate outlier.

Category	Count
Below IQR (Low Outliers)	0
Above IQR (High Outliers)	179
Normal Range	1524

(Table 2: Rate based Outliers)

The rate-based outlier analysis highlights 179 products (approximately 10.5%) whose average purchase prices exceed the upper statistical threshold of ₹800. These high-rate items represent premium or specialized surgical supplies whose prices vary significantly across

vendors. No low outliers were observed because the lower bound lies below zero, which is not applicable to rate data. Identifying these high-price items is essential for cost optimization, negotiation strategies, and planning bulk purchases during low-price months.

2.2.1.3 Inventory Classification-

To prioritise inventory control, a two-stage classification was applied:

1. ABC Analysis-

ABC analysis was applied to the most recent year (April 2024–March 2025) to prioritise items by their contribution to total purchase value. Items were ranked by purchase amount and categorised as:

- **A:** top 20% of items contributing around 80% of total value,
- **B:** next 30% contributing about 15%,
- **C:** remaining 50% contributing the final 5%.

This classification highlights where management should focus attention and working capital. ABC analysis is particularly appropriate here because it simplifies decision-making for a large product base (over 1,600 items) and directly links to the overstocking problem—ensuring high-value products are neither over-purchased nor understocked.

2. VED Classification-

The A-category items were further classified using VED analysis—a method that categorizes products based on their operational criticality:

- **V (Vital):** items essential for daily operations,
- **E (Essential):** necessary but not immediately critical,
- **D (Desirable):** low-criticality or optional items.

Tagging was done using product names and expert input from the organization.

The ABC–VED combination is valuable because it addresses both financial value (ABC) and functional importance (VED). For example, an AV (A-Vital) item like *Infusion Sets* demands continuous availability, while an AD (A-Desirable) item like *Abdominal Belt* can be restocked less frequently. This matrix directly informs stocking and procurement priorities.

2.2.1.4 Trend and Seasonality Analysis-

The three-year data (April 2022–March 2025) have been analysed to identify seasonal and cyclical purchasing patterns. Monthly aggregations are used to observe demand fluctuations and purchasing trends. Detecting seasonal peaks and troughs will allow the company to plan purchases more effectively, reducing last-minute orders and ensuring stock availability during high-demand periods.

2.2.1.5 Demand Pattern Classification

The 58 AV items were further analysed to understand their consumption behaviour.

Two statistical indicators were used:

- Coefficient of Variation (CV) = Standard Deviation / Mean Quantity
- Zero-Months % = (Number of months with zero purchase ÷ Total months) × 100

Based on these metrics:

- Items with low CV and fewer zero months were labelled “Regular”.
- Items with high CV or long zero-month gaps were labelled “Intermittent”.

2.2.1.6 Forecasting Models

This distinction guided the choice of forecasting models suitable for each product type.

Demand Type	Model Used	Description	Reason for Usage
Regular Items (13)	Holt–Winters Exponential Smoothing	Captures both trend and seasonality through three parameters — α (level), β (trend), and γ (seasonality).	Appropriate for continuous, moderately seasonal demand.
Intermittent Items (45)	Croston’s Method	Separates non-zero demand intervals and estimates both demand size and frequency.	Designed for sparse or zero-heavy demand patterns.

(Table 3 - Methods used for Forecasting and their justification)

1. Holt–Winters Exponential Smoothing (for Regular-Demand Items)

Purpose:

Holt–Winters Exponential Smoothing is a time series forecasting method that captures trend and seasonality in regularly observed data. It extends simple exponential smoothing by introducing components for level, trend, and seasonal effects, making it ideal for products with continuous, stable demand.

Concept:

The model updates three components recursively for each time period t :

$$\begin{aligned} \text{Level } (L_t) : L_t &= \alpha \frac{Y_t}{S_{t-m}} + (1 - \alpha)(L_{t-1} + T_{t-1}) \\ \text{Trend } (T_t) : T_t &= \beta(L_t - L_{t-1}) + (1 - \beta)T_{t-1} \\ \text{Seasonal } (S_t) : S_t &= \gamma \frac{Y_t}{L_t} + (1 - \gamma)S_{t-m} \end{aligned}$$

Where:

- Y_t : Actual demand at time t
- L_t : Smoothed estimate of the series' mean (level)
- T_t : Smoothed estimate of the trend
- S_t : Seasonal component
- m : Number of periods in one season (e.g., $m = 12$ for annual seasonality)
- α, β, γ : Smoothing parameters, $0 \leq \alpha, \beta, \gamma \leq 1$

Forecast Equation:

$$\hat{Y}_{t+h} = (L_t + hT_t)S_{t-m+h}$$

Interpretation:

- Suitable for steady, continuous demand (e.g., RMS INFUSION SET).
- Generates smooth forecasts reflecting gradual demand evolution.
- Parameters (α, β, γ) control sensitivity — higher values react faster to new trends.

Why it fits Rajesh Surgical's data:

Regular items like IV sets and syringes exhibit consistent monthly usage; Holt–Winters captures their mild trend while smoothing short-term fluctuations.

2. Croston's Method (for Intermittent-Demand Items)

Purpose:

Croston's Method is specifically designed for intermittent or lumpy demand — where many time periods have zero demand, punctuated by occasional non-zero orders. Traditional models like Holt–Winters or ARIMA perform poorly here, as they assume continuous demand.

Concept:

Instead of forecasting every period's demand, Croston separates the problem into two components:

- The demand size z_t when demand occurs, and
- The interval p_t between non-zero demands.

Each component is updated independently using exponential smoothing:

$$\begin{aligned} \text{Demand Size } (z_t) : z_t &= \alpha y_t + (1 - \alpha)z_{t-1} \\ \text{Interval } (p_t) : p_t &= \beta q_t + (1 - \beta)p_{t-1} \end{aligned}$$

where:

- y_t : actual demand in period ttt (zero or positive)
- q_t : interval (number of periods since last non-zero demand)
- α, β : smoothing constants

Forecast Equation:

$$\hat{y}_{t+1} = \frac{z_t}{p_t}$$

Interpretation:

- Produces an average expected demand per period (not specific months).
- Works best for products with sporadic orders, e.g., OMRON NEBULIZER.
- Doesn't overestimate zero-demand months, avoiding unnecessary stock.

Enhancement in this project:

Since Croston gives an overall monthly average, the forecasts were post-processed by redistributing quantities across months based on historical non-zero demand probability, resulting in realistic, time-aware projections.

Why it fits Rajesh Surgical's data:

Many AV items are ordered irregularly due to hospital tenders or patient-specific needs. Croston's method handles this discontinuity while still giving a stable annual demand estimate.

Forecasts were validated visually for consistency and aggregated to produce a 12-month demand projection across all AV items.

The forecasting process for intermittent-demand items using the Croston method initially produced a constant expected monthly quantity, as this model predicts average demand size and frequency rather than the specific timing of purchases. To make the results more business-realistic, the forecasts were post-processed to reflect actual purchasing behaviour.

For each product, the historical data was analyzed to determine the probability of a non-zero purchase month. The forecasted annual quantity was then redistributed across the 12 forecast months according to these probabilities, ensuring that months with historically higher likelihoods of ordering received proportionally higher forecasted quantities, while zero-purchase months remained at or near zero.

This adjustment preserved Croston's total annual forecast but provided a more accurate month-by-month projection aligned with Rajesh Surgical's intermittent purchasing patterns.

A regular-product forecast file and an intermittent-product forecast file were created, and later a combined forecast workbook with monthly predictions for all 58 AV items was created. These results provide Rajesh Surgical with an actionable view of expected demand for the upcoming year, supporting proactive procurement planning and stock control.

Let's move on to Problem Statement-2:

2.2.2 Problem Statement 2: Inefficient Cost Management

2.2.2.1 Objective-

To analyse purchasing patterns for cost inefficiencies caused by supplier price fluctuations and identify opportunities for bulk buying or improved procurement timing.

2.2.2.2 Data Scope and Filtering-

For this analysis, purchase data from April 2024 to March 2025 were extracted from the cleaned AV dataset to represent the most recent fiscal year. The focus remained on the 58 AV items, covering variables: *Month, Product Name, Quantity, Average Rate*.

2.2.2.3 Price Variability Analysis-

To evaluate cost fluctuations, price variability metrics were computed per item:

- Mean_Rate – average monthly purchase price
- Std_Rate – standard deviation of monthly rate
- CV_Rate = Std_Rate / Mean_Rate

Items with CV_Rate > 0.3 were considered highly volatile (while some Items with CV_Rate > 0.1 were also volatile), indicating inconsistent vendor pricing or irregular market rates.

The selected volatile products were identified for further trend visualisation. We further visualised the selected Price-Volatile Items and created individual rate trend charts showing price peaks and troughs. A summary of potential cost savings for selected AV items was created. These insights inform data-driven purchasing decisions and supplier negotiations, reducing procurement costs while maintaining stock availability

3. Results & Findings

This section presents the outcomes of the analyses conducted under the two primary problem statements:

- Inventory Planning & Suboptimal Purchase Strategy, and
- Inefficient Cost Management

The results combine quantitative evidence, statistical observations, and visual representations to demonstrate how the proposed framework can improve purchasing and stock control at Rajesh Surgical.

3.1 Problem Statement 1: Inventory Planning & Suboptimal Purchase Strategy

3.1.1 ABC Analysis

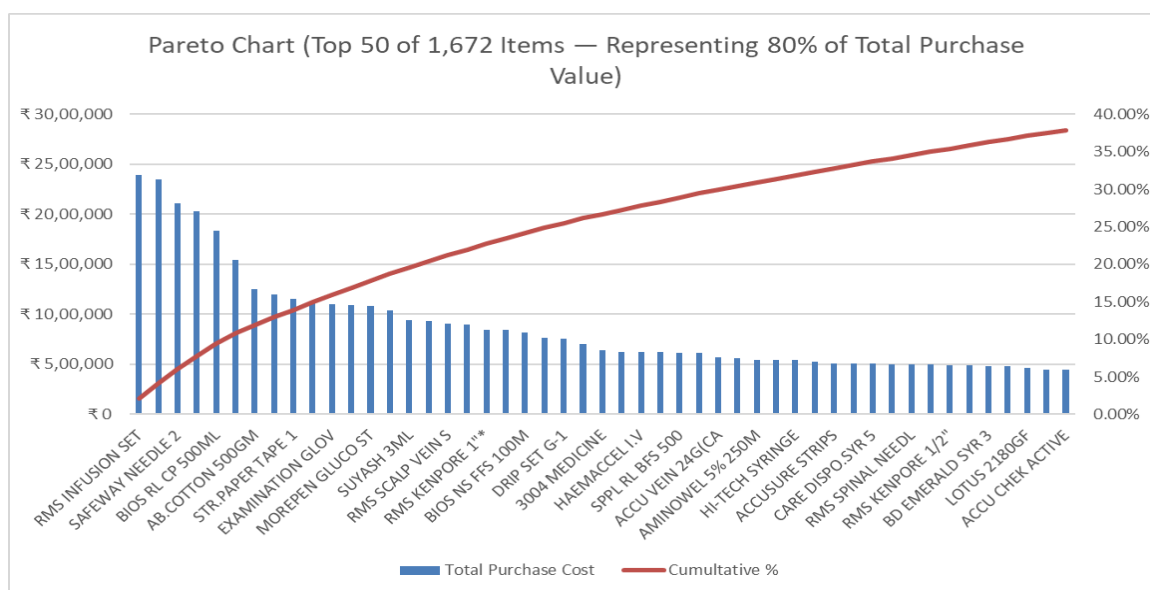
Using FY 2024–25 data, an ABC classification was conducted to categorize inventory based on total purchase value:

Category	No. of Items	% of Total Items	Total Purchase	Purchase %
A	327	20%	₹ 9,08,08,953	80%
B	448	27%	₹ 1,70,95,107	15%
C	896	54%	₹ 56,95,223	5%
Total	1671	100%	₹ 11,35,99,284	100%

(Fig 3 - Items Classification into A, B, C Category)

This follows the Pareto principle, where 20% of items account for 80% of total purchases.

This result identifies the high-impact “A” items that should receive focused attention in procurement and inventory control to optimize cost and availability.



(Fig 4 - Pareto Chart of Top 50 Items)

3.1.2 VED Classification

To complement financial prioritization (ABC analysis) with operational importance, VED classification was applied to the 327 “A-category” items, identifying which are *Vital*, *Essential*, or *Desirable* for hospital operations.

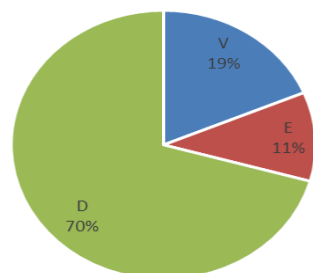
Category	V	E	D	Grand Total
A	61	35	231	327
B	44	33	371	448
C	27	34	835	896
Grand Total	132	102	1437	1671

(Fig 5 - ABC-VED Matrix for all 1671 Items)

Category	No. of Items	% of Total Items	Total purchase per Category	Purchase Share out of A	Purchase Share out of Total
V	61	19%	₹ 2,41,17,584	27%	21%
E	35	11%	₹ 1,02,48,520	11%	9%
D	231	71%	₹ 5,64,42,849	62%	50%
Total	327	100%	₹ 9,08,08,953	100%	80%

(Fig 6 - Classification of A Category Items in V, E, D Category)

Percentage (%) Share of VED Items



(Fig 7 - Percentage Share of V, E, & D Category Items within the A Category)

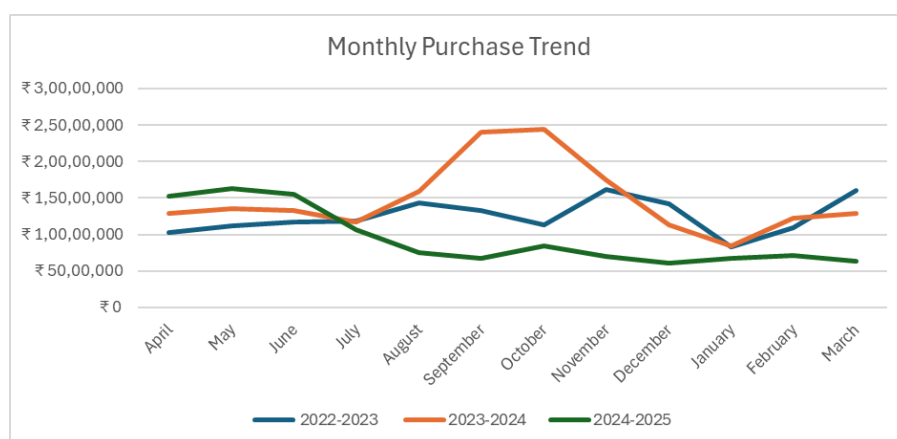
The Vital (V) items, though only 19% of A-category items, contribute 27% of A purchases and 21% of total spending, covering essential hospital supplies like infusion sets, syringes, and IV fluids that require uninterrupted availability.

The Essential (E) group forms 11% of A items and 9% of total spending, including frequently used items such as cotton and dressing pads.

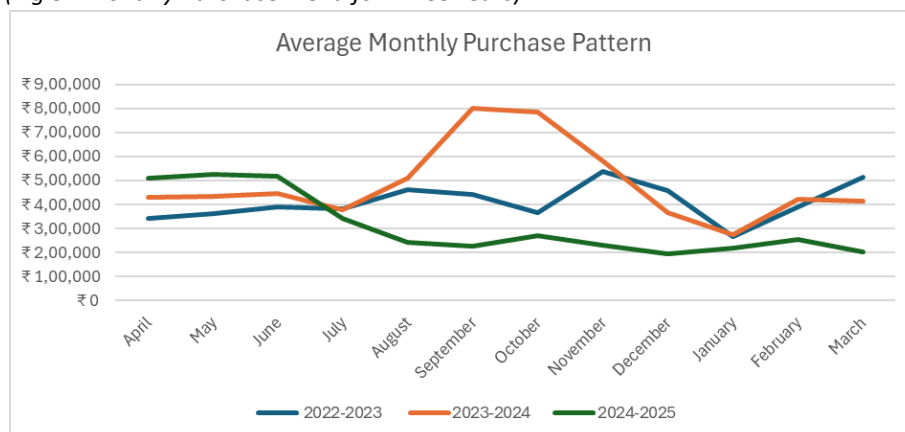
The Desirable (D) items make up the majority (71%) and account for half of total purchase value, indicating a large number of low-priority consumables consuming significant expenditure.

Overall, the findings suggest that A–V items need tighter monitoring and vendor agreements to ensure availability.

3.1.3 Trend and Seasonality Analysis



(Fig 8 - Monthly Purchase Trend for Three Years)



(Fig 9 - Average Monthly Purchase Trend for Three Years)

To explore purchasing behavior over time, monthly purchase totals were aggregated across three financial years (FY 2022–23, FY 2023–24, and FY 2024–25). Two charts are developed to visualize trends and seasonality.

Across FY 2022–23 to 2024–25, monthly purchase trends show clear seasonality.

- FY 2022–23 remained stable with mild peaks in November and March.
- FY 2023–24 exhibited a sharp rise during August–October, followed by a post-festive dip.
- FY 2024–25 reflected lower yet steady purchases.

The average monthly pattern across years confirms cyclical demand, with consistent spikes in Q2–Q3 (Aug–Oct) and March.

These insights indicate predictable seasonal behavior, supporting strategic bulk purchasing and rate optimization ahead of high-demand periods.

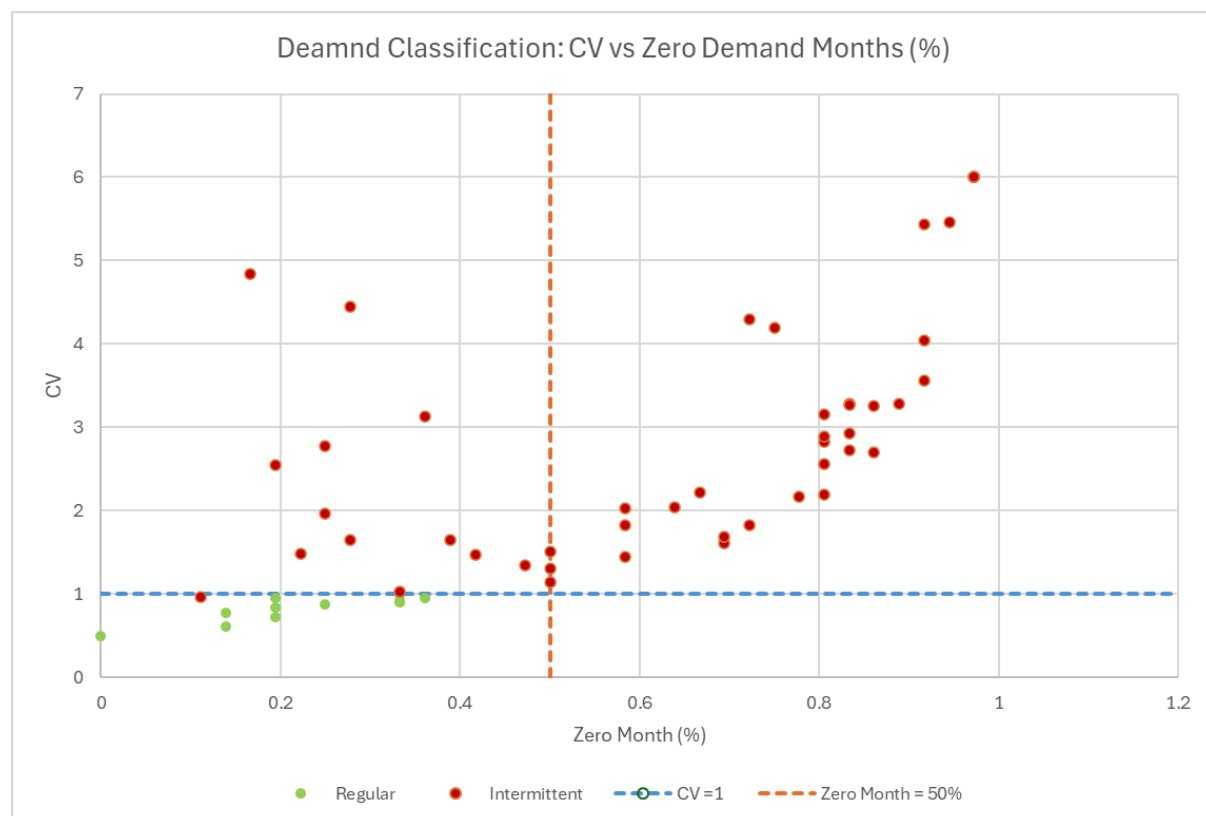
3.1.4 Demand Classification (Regular vs Intermittent)

The initial list of AV (A–Vital) items contained 61 products, but after consultation with the business owner, a few items with slightly different names but identical specifications were found to be duplicates (for example, alternate spellings or package-size variations of the same product). These were consolidated to ensure data accuracy and consistency, resulting in a final list of 58 unique AV items for further analysis.

The 58 AV items were further classified based on their demand variability and purchasing frequency. Two indicators—Coefficient of Variation (CV) and Zero-Months Percentage—were used to segment the items.

Demand Type	No. of Items	Criteria
Regular	13 (22%)	$CV \leq 1$ or Zero-Months % $\leq 50\%$
Intermittent	45 (78%)	$CV > 1$ or Zero-Months % $> 50\%$

(Table 4 - Demand Pattern Summary)



(Fig 10 - Scatter Plot of CV vs Zero-Months Percentage)

The scatter plot clearly illustrates two distinct clusters — one representing steady, regular-demand items, and the other comprising items with sporadic or irregular purchase patterns. This classification guided the choice of forecasting models for each group, using Holt-Winters for regular items and Croston’s method for intermittent ones.

3.1.5 Forecasting Results

Forecasting was carried out for April 2025 – March 2026 using two methods:

- Holt-Winters exponential smoothing for *regular-demand items*
- Croston’s method (with probability-based redistribution) for *intermittent-demand items*

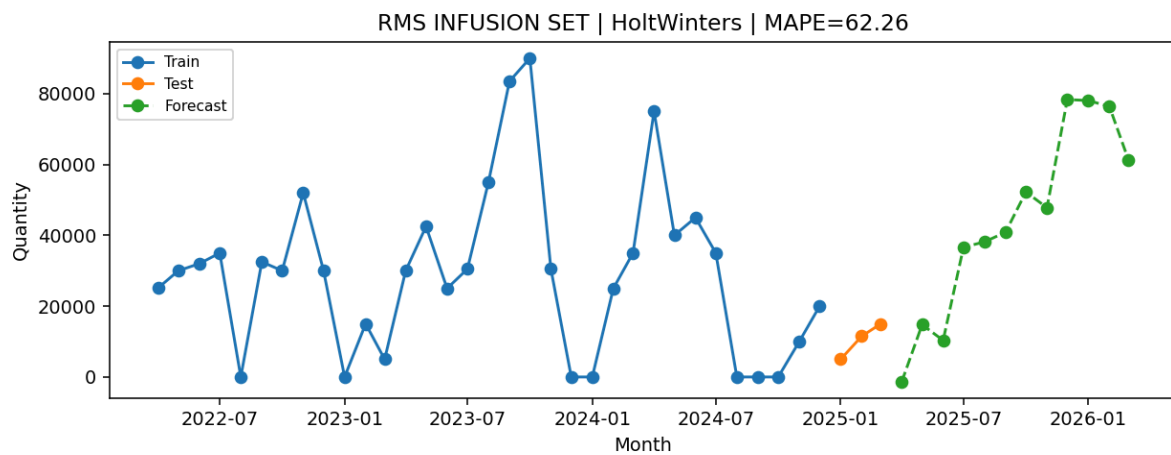
1. Regular-Demand Items (Holt-Winters Model)

These items exhibited consistent monthly consumption with mild seasonality.

Example: *RMS INFUSION SET* showed smooth forecast continuity with minor variance.

Product_Name	Forecast_Month	Forecast_Quantity	Avg_Rate_used	Forecast_Amount	Model	Demand_Type
RMS INFUSION SET	2025-04-01	-1357	₹ 9.20	-₹ 12,477.13	HoltWinters	Regular
RMS INFUSION SET	2025-05-01	14773	₹ 9.20	₹ 1,35,864.35	HoltWinters	Regular
RMS INFUSION SET	2025-06-01	10322	₹ 9.20	₹ 94,933.31	HoltWinters	Regular
RMS INFUSION SET	2025-07-01	36635	₹ 9.20	₹ 3,36,924.02	HoltWinters	Regular
RMS INFUSION SET	2025-08-01	38147	₹ 9.20	₹ 3,50,835.14	HoltWinters	Regular
RMS INFUSION SET	2025-09-01	40870	₹ 9.20	₹ 3,75,878.14	HoltWinters	Regular
RMS INFUSION SET	2025-10-01	52233	₹ 9.20	₹ 4,80,383.13	HoltWinters	Regular
RMS INFUSION SET	2025-11-01	47796	₹ 9.20	₹ 4,39,570.88	HoltWinters	Regular
RMS INFUSION SET	2025-12-01	78333	₹ 9.20	₹ 7,20,415.55	HoltWinters	Regular
RMS INFUSION SET	2026-01-01	78050	₹ 9.20	₹ 7,17,812.47	HoltWinters	Regular
RMS INFUSION SET	2026-02-01	76369	₹ 9.20	₹ 7,02,358.41	HoltWinters	Regular
RMS INFUSION SET	2026-03-01	61349	₹ 9.20	₹ 5,64,221.21	HoltWinters	Regular

(Fig 11 - Forecasting Result for “RMS INFUSION SET” using Holt Winter’s Method)



(Fig 12 - Forecasting Trend for “RMS INFUSION SET” using Holt Winter’s Method)

The chart shows the training data (blue), test validation period (orange), and the forecasted demand (green) for the upcoming year. The gradual upward slope of the forecast line suggests an expected increase in demand, supporting the need for timely restocking and advance procurement planning.

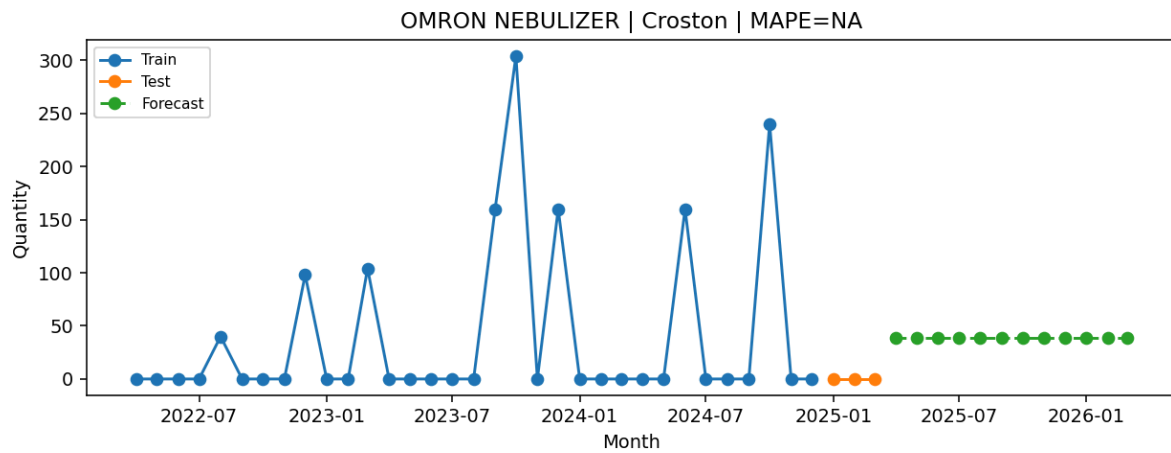
2. Intermittent-Demand Items (Croston’s Method with Redistribution)

Items with sporadic purchasing patterns were modeled using the Croston method, which estimates average demand size and interval frequency. Croston’s model initially produced a uniform forecast across all months due to its probabilistic design for zero-heavy demand.

Example- OMRON NEBULIZER

Product_Name	Forecast_Mon	Forecast_Quanti	Avg_Rate_us	Forecast_Amou	Model	Demand_Ty
OMRON NEBULIZER	2025-04-01	39	₹ 1,060.00	₹ 41,374.22	Croston	Intermittent
OMRON NEBULIZER	2025-05-01	39	₹ 1,060.00	₹ 41,374.22	Croston	Intermittent
OMRON NEBULIZER	2025-06-01	39	₹ 1,060.00	₹ 41,374.22	Croston	Intermittent
OMRON NEBULIZER	2025-07-01	39	₹ 1,060.00	₹ 41,374.22	Croston	Intermittent
OMRON NEBULIZER	2025-08-01	39	₹ 1,060.00	₹ 41,374.22	Croston	Intermittent
OMRON NEBULIZER	2025-09-01	39	₹ 1,060.00	₹ 41,374.22	Croston	Intermittent
OMRON NEBULIZER	2025-10-01	39	₹ 1,060.00	₹ 41,374.22	Croston	Intermittent
OMRON NEBULIZER	2025-11-01	39	₹ 1,060.00	₹ 41,374.22	Croston	Intermittent
OMRON NEBULIZER	2025-12-01	39	₹ 1,060.00	₹ 41,374.22	Croston	Intermittent
OMRON NEBULIZER	2026-01-01	39	₹ 1,060.00	₹ 41,374.22	Croston	Intermittent
OMRON NEBULIZER	2026-02-01	39	₹ 1,060.00	₹ 41,374.22	Croston	Intermittent
OMRON NEBULIZER	2026-03-01	39	₹ 1,060.00	₹ 41,374.22	Croston	Intermittent

(Fig 13 - Forecasting Result for “OMRON NEBULIZER” using Croston Method)



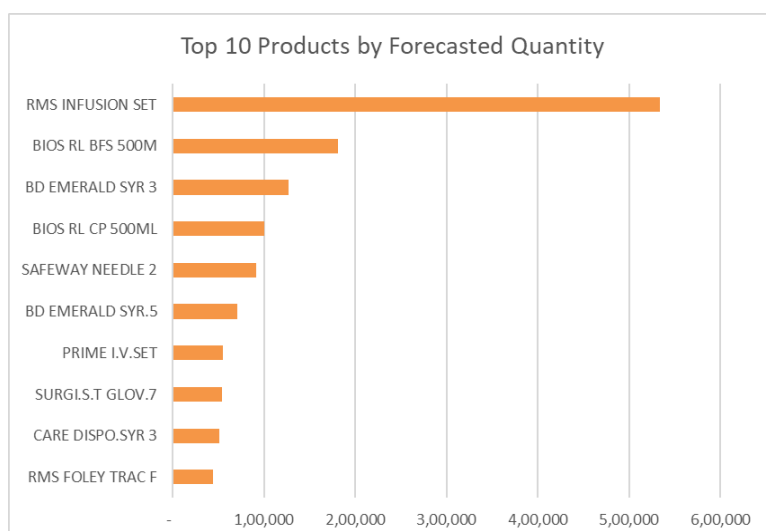
(Fig 14- Forecasting Trend for “OMRON NEBULIZER” using Croston Method)

The figure illustrates intermittent purchase behavior, where long periods of zero demand are followed by sudden order spikes. The Croston-based forecast (green) provides smoothed predictions of expected purchasing months without overestimating demand in zero periods. This pattern suggests that the product should not be stocked continuously but rather ordered in planned intervals based on procurement lead time and supplier availability.

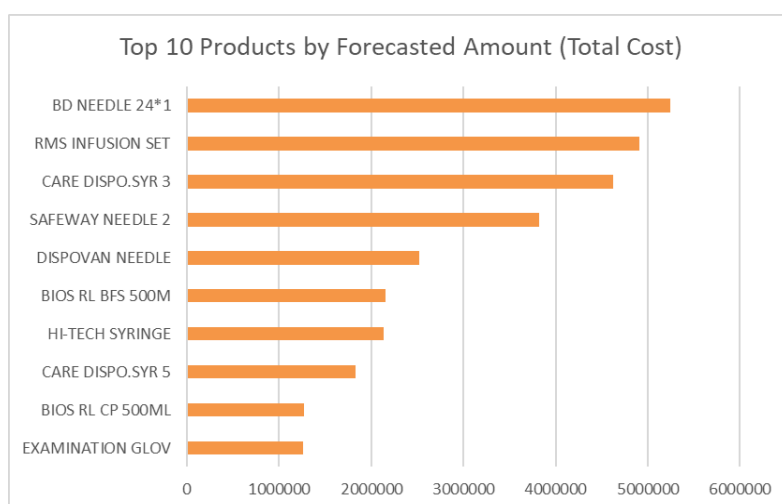
To enhance realism, the forecasted annual quantity was redistributed across 12 months based on each item’s historical probability of non-zero purchases, ensuring that forecasted demand aligns with actual ordering behavior.

Product_Name	Forecast_Month	Post_Forecast_Quant	Avg_Rate_us	Post_Forecast_Amou
OMRON NEBULIZER	2025-04-01	0	1060	0
OMRON NEBULIZER	2025-05-01	0	1060	0
OMRON NEBULIZER	2025-06-01	156	1060	165360
OMRON NEBULIZER	2025-07-01	0	1060	0
OMRON NEBULIZER	2025-08-01	0	1060	0
OMRON NEBULIZER	2025-09-01	0	1060	0
OMRON NEBULIZER	2025-10-01	156	1060	165360
OMRON NEBULIZER	2025-11-01	0	1060	0
OMRON NEBULIZER	2025-12-01	156	1060	165360
OMRON NEBULIZER	2026-01-01	0	1060	0
OMRON NEBULIZER	2026-02-01	0	1060	0
OMRON NEBULIZER	2026-03-01	0	1060	0

(Fig 15 - Forecasting Result for “OMRON NEBULIZER” after redistribution based on historical probability)



(Fig 16 - Top 10 products by Forecasted Quantity)



(Fig 17 - Top 10 products by Forecasted Amount)

The two charts above highlight the forecasted demand and corresponding procurement cost distribution for the upcoming fiscal year.

- RMS INFUSION SET dominates the forecasted quantity chart with projected demand exceeding 500,000 units, underscoring its status as a high-consumption, high-turnover product.
- Other frequently purchased consumables such as BIOS RL BFS 500ML, BD EMERALD SYR 3ML, and SAFEWAY NEEDLE 2 also appear prominently, reflecting steady recurring demand across hospitals.
- When viewed in terms of forecasted total procurement cost, products like BD NEEDLE 24*1, RMS INFUSION SET, and CARE DISPO SYR 3ML emerge as top spend contributors.
- These findings reveal that a small subset of products (top 10 items) account for a significant share of total forecasted volume and cost, aligning with the Pareto principle observed in the earlier ABC analysis.

NOTE: For Monthly Forecasting Result of all 58 AV Items, refer to the folder link & table given in the Appendix at the end of the report.

3.2 Problem Statement 2: Inefficient Cost Management

3.2.1 Descriptive Insights

Descriptive analysis of the purchase data revealed high variability in purchase quantity, rate, and frequency across different items, suggesting inconsistent procurement patterns.

To illustrate this, an item-level examination was performed on *RMS Infusion Set* — a high-value and frequently purchased medical supply. Between April 2024 and March 2025, this item was purchased 20 times, with quantities ranging from 500 to 49,500 units per order. The purchase rate fluctuated between ₹9.15 and ₹11.33, reflecting a price variation of nearly 24% for the same product.

Date	Item Description	Quantity	Rate	Amount
06-06-2024	RMS INFUSION SET	20000	9.1512	183024
01-07-2024	RMS INFUSION SET	25000	9.1512	228780
24-07-2024	RMS INFUSION SET	10000	9.1969	91968.7
06-11-2024	RMS INFUSION SET	10000	9.1969	91968.7
02-12-2024	RMS INFUSION SET	5000	9.1969	45984.35
13-12-2024	RMS INFUSION SET	15000	9.1969	137953.05
23-01-2025	RMS INFUSION SET	5000	9.1969	45984.35
06-02-2025	RMS INFUSION SET	10000	9.1969	91968.7
03-03-2025	RMS INFUSION SET	15000	9.1969	137953.05
08-04-2024	RMS INFUSION SET	5000	9.2496	46248
10-04-2024	RMS INFUSION SET	20000	9.348	186960
15-04-2024	RMS INFUSION SET	500	9.456	4728
15-04-2024	RMS INFUSION SET	49500	9.456	468072
04-05-2024	RMS INFUSION SET	8000	9.456	75648
04-05-2024	RMS INFUSION SET	4000	9.456	37824
04-05-2024	RMS INFUSION SET	1000	9.456	9456
04-05-2024	RMS INFUSION SET	2000	9.456	18912
07-05-2024	RMS INFUSION SET	15000	9.456	141840
07-05-2024	RMS INFUSION SET	10000	9.456	94560
14-06-2024	RMS INFUSION SET	25000	9.456	236400
03-02-2025	RMS INFUSION SET	1500	11.33	16995

(Fig 18 - Price (Rate) Variability in “RMS INFUSION SET” after redistribution based on historical probability)

Such inconsistency in both order size and pricing indicates the absence of a structured reorder policy or long-term vendor rate contract. The day-wise transaction pattern further confirms irregular purchase intervals, showing that procurement decisions are reactive rather than forecast-based.

Overall, these descriptive findings highlight the need for a data-driven purchase planning approach to ensure stable supply, price consistency, and optimal inventory levels.

3.2.2 Price Variability Analysis

Price variability was examined for all 58 AV items between April 2024 and March 2025.

Each item's Mean Rate, Min Rate, Max Rate, and Coefficient of Variation (CV_Rate) were computed.

Product	Mean_Rate	Max_Rate	Min_Rate	StdDev_Rate	CV_Rate
HI-TECH SYRINGE	133.1376943	254.4766667	100.8976923	57.20820207	0.429691999
DRIP SET G-1	2.583391664	3.7	1.6965	0.935012606	0.361932191
BD EMERALD SYR 3	3.416219415	7.092105263	3	1.161087828	0.339875074
DISPOVAN NEEDLE	83.21715624	143.5	64	24.30082568	0.292017016
ACCU CHEK ACTIVE	927.0392902	1410	700.66	197.70725	0.21326739
EXAMINATION GLOV	108.7848925	135.1481481	84	20.94495926	0.192535551
BPL NEBULIZER BR	840.001875	960	720	125.3339539	0.149206755
OXYGEN CONCENTRA	27827.49833	33928.57	23215	3888.614999	0.139740014
FLAM.WHEEL CHAIR	4259.65	4812.7	3700	513.8077718	0.120622063
I.V.STAND	354.5474359	410	300	40.66912586	0.11470715
GB DISPO.SYRINGS	2.5875	2.75	2.1	0.293973561	0.11361297

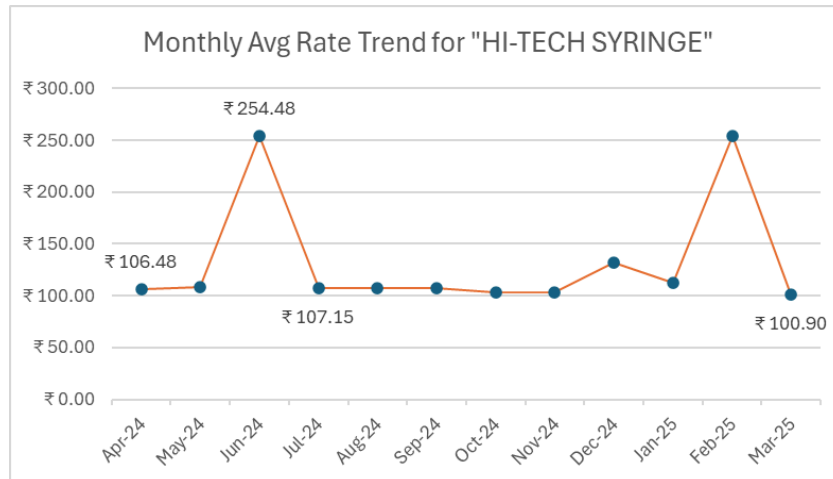
(Fig 19 - Mean Rate, Min Rate, Max Rate, and CV for some of the Items)

3.2.3 Optimal Purchase Timing and Cost-Saving Potential

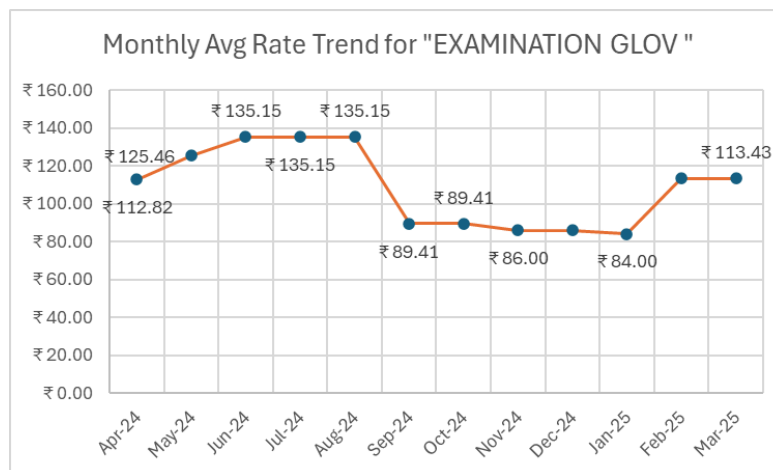
For each volatile item, the lowest average price month(s) were identified, and potential saving percentage was calculated:

$$\text{Potential Saving (\%)} = \frac{(\text{Mean Rate} - \text{Min Rate})}{\text{Mean Rate}} \times 100$$

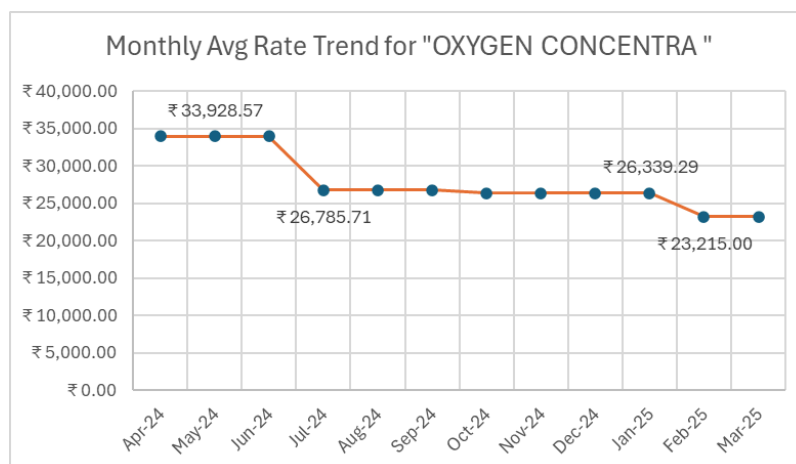
Here are trend charts of the selected 4 Items and their summary table:



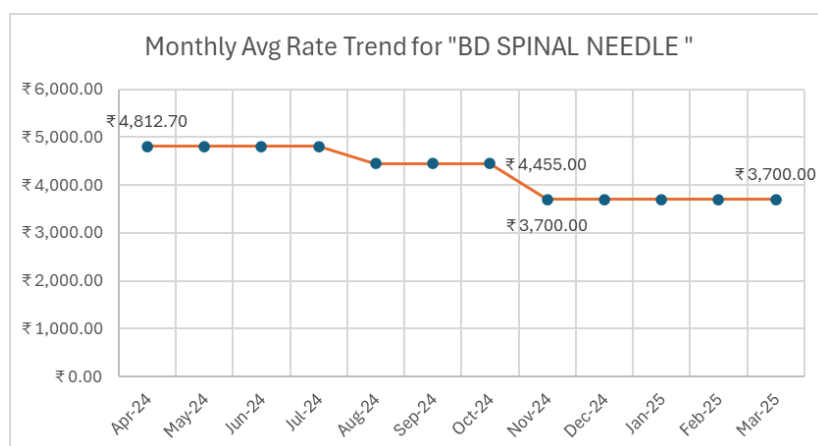
(Fig 20 - Monthly Avg Rate Trend for "HI-TECH SYRINGE")



(Fig 21 - Monthly Avg Rate Trend for "EXAMINATION GLOV")



(Fig 22 - Monthly Avg Rate Trend for "OXYGEN CONCENTRA")



(Fig 23 - Monthly Avg Rate Trend for "BD SPINAL NEEDLE")

Product Name	Mean Rate	Max Rate	Min Rate	CV Rate	Recommended Buy Month	Potential Saving %
HI-TECH SYRINGE	133.137694	254.476667	100.897692	0.430	Mar	31.95%
EXAMINATION GLOV	108.784892	135.148148	84	0.193	Jan	29.51%
OXYGEN CONCENTRA	27827.4983	33928.57	23215	0.140	Feb-Mar	19.87%
FLAM.WHEEL CHAIR	4259.65	4812.7	3700	0.121	Nov-Mar	15.13%

(Fig 24 - Summary of these 4 Products showing Recommended buy Months)

The analysis of price trends for the selected four representative AV items revealed clear evidence of seasonal or supplier-driven price variation.

- HI-TECH SYRINGE exhibited the highest variability (CV = 0.43), with prices peaking mid-year and bottoming in March, suggesting significant scope for cost reduction (~32%) if procurement is planned during that low-rate period.
- EXAMINATION GLOV showed a similar pattern, with prices stabilizing around January, offering an estimated 29.5% saving potential.
- OXYGEN CONCENTRA, being a high-value equipment item, displayed moderate variability but still offered nearly 20% savings when purchased between February and March.
- FLAM.WHEEL CHAIR had relatively stable pricing, yet strategic buying during November–March could achieve ~15% savings.

These representative cases demonstrate that even within the AV category, procurement cost optimization opportunities vary by product type. By aligning purchase orders with low-price months, Rajesh Surgical can realise tangible savings while ensuring an uninterrupted supply of critical medical products.

NOTE: For further understanding of Price Variability analysis & Optimal Purchase Timing of all 58 AV Items, refer to the folder link & table given in the Appendix at the end of the report.

4. Interpretation of Results and Recommendations

This section consolidates insights from the analyses on inventory management and purchase optimization. The findings demonstrate how Rajesh Surgical can enhance its operational efficiency, minimize procurement costs, and implement data-driven decision-making practices.

4.1 Interpretation of Results

4.1.1 Problem Statement 1 – Inventory Planning & Suboptimal Purchase Strategy

The analysis of three years of purchase data revealed clear patterns in product demand and inventory behavior.

1. **Inventory Prioritization (ABC–VED):**

The ABC–VED classification highlighted that a small subset of items (Category A–Vital) represents a large share of procurement expenditure and operational importance. These 58 AV items form the backbone of Rajesh Surgical's business and therefore require continuous stock monitoring and careful reorder planning.

2. **Demand Behavior Segmentation:**

Using statistical measures such as Coefficient of Variation (CV) and Zero-Months Percentage, the 58 AV items were segmented into 13 regular and 45 intermittent products. This differentiation clarified that most critical items have irregular consumption, making uniform forecasting or replenishment impractical.

3. **Forecasting Insights:**

- The Holt-Winters exponential smoothing model effectively captured monthly trends for regular-demand items, producing stable, interpretable forecasts.
- The Croston method, enhanced through probability-based redistribution, yielded realistic month-wise forecasts for intermittent items, maintaining alignment with their historical purchase frequencies.

Together, these forecasts provide Rajesh Surgical with a 12-month demand projection for April 2025–March 2026, enabling proactive purchase planning and better vendor coordination.

4. **Operational Implication:**

The demand forecasts indicate a moderate upward trend for several recurring consumables such as RMS INFUSION SET and SAFEWAY SYR 3ML. This suggests a potential rise in hospital consumption or order frequency, highlighting the need to maintain sufficient working capital and supplier readiness for these products.

4.1.2 Problem Statement 2 – Inefficient Cost Management

1. **Price Variability:**

Analysis of unit rates between April 2024 and March 2025 identified significant price fluctuations across certain AV items. Products such as HI-TECH SYRINGE and EXAMINATION GLOV showed high coefficients of variation ($CV > 0.3$), indicating volatile vendor pricing.

2. **Procurement Timing Opportunities:**

By comparing mean and minimum rates, clear seasonal price troughs were identified. Strategic purchasing during these months (e.g., March for HI-TECH SYRINGE, January for EXAMINATION GLOV) could yield 15–30% savings for selected items without compromising service continuity.

3. Business Interpretation:

The findings emphasize that price trends are not random but follow patterns influenced by supply cycles, vendor negotiations, and fiscal-year-end discounts. By aligning purchase timing with these low-rate windows, Rajesh Surgical can achieve substantial cost optimization across high-spend products.

4.2 Recommendations

Area	Recommendation	Expected Outcome
Inventory Planning	Implement separate forecasting models for regular (Holt-Winters) and intermittent (Croston) items.	Improved demand accuracy and reduced stockouts.
Procurement Scheduling	Align orders for volatile items with identified low-price months.	15–30% cost savings on select AV items.
Vendor Negotiation	Use historical rate variability as leverage for rate contracts and volume discounts.	More stable pricing and stronger supplier relationships.
Working Capital Management	Allocate higher capital for vital, high-consumption items; maintain reorder points for intermittent ones.	Optimal cash flow utilization.
Automation & Monitoring	Integrate the forecasting and price monitoring system into Excel or ERP dashboards for real-time updates.	Continuous data-driven decision-making.

4.3 Strategic Implications

- The combination of forecasting and price analysis establishes a holistic procurement intelligence framework.
- Data-driven planning reduces dependency on ad-hoc decision-making and enhances transparency.
- Implementing the proposed recommendations can reduce carrying costs, avoid emergency purchases, and improve profit margins through informed bulk buying.
- Over time, consistent application of this analytical framework can evolve into a fully automated procurement system, supported by dashboards and predictive alerts.

5. Appendix – Links to Analysis Files and Code Files

All analysis outputs, cleaned datasets, and supporting code files used in this project have been compiled in a dedicated Google Drive folder called “Analysis” for evaluation and reproducibility. The folder contains both Excel-based summaries and Python scripts used for data processing, forecasting, and visualisation.

Link to Folder - [“Analysis”](#). Here’s a Description of all the files/folders within this “Analysis” folder:

File / Folder Name	Description / Purpose
1. All Three Cleaned Datasets	This sub-folder within the "Analysis" folder has all three cleaned datasets (<i>"project_data_202404_202503_cleaned.xlsx"</i> , <i>"project_data_202304_202403_cleaned.xlsx"</i> , <i>"project_data_202204_202303_cleaned.xlsx"</i>) and the data cleaning Python code (<i>"data_cleaning_code.ipynb"</i>).
2. Forecasting	This sub-folder within the "Analysis" folder has code used for forecasting (<i>"Forecasting_Code.ipynb"</i>), a forecasting Excel file (<i>"AV_forecast_combined_final.xlsx"</i>) containing monthly forecasting for all 58 AV products. It also has a subfolder called <i>"forecasting plots"</i> , which has forecasting trends for all 58 AV Products.
3. Analysis_202204_202503_data.xlsx	Contains all the analyses conducted on the entire three-year data, including Trend, Seasonality, etc.
4. Analysis_202404_202503_data.xlsx	Contains all the Analysis conducted on the data of the period Apr'2024-Mar'2025, including ABC-VED Analysis, Descriptive Stats, Charts, Trends, etc.
5. Analysis_AV_forecast_combined_final.xlsx	This file contains the Monthly forecasting of all 58 AV products and all the analyses done on forecasting- the tables, charts, trends, etc.
6. Analysis_for_Problem_Statement-2.xlsx	This file contains all the analysis done for Problem Statement-2, like price variability Analysis of 58 AV Products

Link to Proof of Originality Docs - [Link](#)